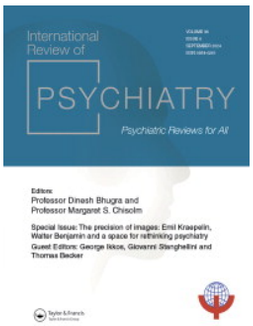

International Review of Psychiatry

ISSN: 0954-0261 (Print) 1369-1627 (Online) Journal homepage: www.tandfonline.com/journals/iirp20

Digital visions: the experience of self and others in the age of the digital revolution

Vittorio Gallese

To cite this article: Vittorio Gallese (2024) Digital visions: the experience of self and others in the age of the digital revolution, International Review of Psychiatry, 36:6, 656-666, DOI: 10.1080/09540261.2024.2355281

To link to this article: https://doi.org/10.1080/09540261.2024.2355281

© 2024 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group

Published online: 20 May 2024.

Submit your article to this journal

Article views: 4303

View related articles

View Crossmark data

Citing articles: 8 View citing articles

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=iirp20

INTERNATIONAL REVIEW OF PSYCHIATRY
2024, VOL. 36, NO. 6, 656–666
https://doi.org/10.1080/09540261.2024.2355281

Taylor & Francis
Taylor & Francis Group

RESEARCH ARTICLE

OPEN ACCESS Check for updates

Digital visions: the experience of self and others in the age of the digital revolution

Vittorio Gallese[a,b]

[a]Department of Medicine & Surgery - Neuroscience Unit, University of Parma, Parma, Italy; [b]Italian Academy for Advanced Studies in America, Columbia University, New York, NY, USA

ABSTRACT
The digital technological revolution shifted the balance of world perceptual experience, increasing exposure to digital content, introducing a new quality to our perceptual experiences. Embodied cognition offers an ideal vantage point to study how digital technologies impact on selves and their social relations for at least two reasons: first, because of the bodily performative character of the relations and interactions these new media evoke; second, because similar brain-body mechanisms ground our relations with both the physical world and its digital mediations. A closer look is taken at the possible effects of digitization on social communication, on politics, as well as on the constitution of the self and its world relations, especially in the context of the ever-increasing amount of time spent online, with a focus on digital natives. As we explore the complexities of the digital age, it is imperative to critically examine the role of digital technologies in shaping social life and political discourse. By understanding the interplay between content, emotional context, delivery methods, and shareability within digital media landscapes, we can develop strategies to mitigate the negative effects of misinformation and promote informed decision-making in our increasingly digital world.

ARTICLE HISTORY
Received 2 May 2024
Accepted 10 May 2024

KEYWORDS
Digital mediascape; digital natives; embodied simulation; intersubjectivity; psychopathology;

Introduction

> The question of whether people's visual impressions are determined only by natural constants, or additionally by historical variables, is at the very leading edge of research. To move an inch closer to an answer is a hard-won advance. (Walter Benjamin, Selected writings, 2002, Vol.3, p. 146)

Walter Benjamin undoubtedly anticipated much of the contemporary debate on the relationship between aesthetics and technology by emphasizing the historical dimension of perception. According to Benjamin, the advent of modern media and dispositives introduced profound changes in the way perception works. In fact, at the opening of chapter three of *The Art Work in the Age of Mechanical Reproduction* he famously wrote: '*During long periods of history, the mode of human sense perception changes with humanity's entire mode of existence. The manner in which human sense perception is organized, the medium in which it is accomplished, is determined not only by nature but by historical circumstances as well*' (Benjamin, 2008, p. 17). As argued by Andrea Pinotti (2018), in this dense passage Benjamin shows that the change of sense perception produced by new media deeply affects humans' existence, modifying their experience and sensibility, as perception is not simply given but 'organized' within a particular medium.

In this paper, I'll do my best to 'move an inch closer to an answer' by proposing some preliminary considerations on the impact of the widespread use of digital technologies on interpersonal relationships and intersubjectivity, framing them as expressions of our biological nature, on which neuroscience has something to say. Capitalizing upon the embodied perspective on the self and social relationships, I submit that our biological nature cannot be fully understood without considering contemporary digital technology and the socio-cultural practices that it feeds, inspires, and conditions. As argued by the philosopher Gilbert Simondon (2001), technology surpasses any narrow utilitarian purpose: as technology produces new relationships

CONTACT Vittorio Gallese vittorio.gallese@unipr.it Department of Medicine & Surgery - Neuroscience Unit, University of Parma, Parma, Italy; Italian Academy for Advanced Studies in America, Columbia University, New York, NY, USA
© 2024 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group
This is an Open Access article distributed under the terms of the Creative Commons Attribution-NonCommercial-NoDerivatives License (http://creativecommons.org/licenses/by-nc-nd/4.0/), which permits non-commercial re-use, distribution, and reproduction in any medium, provided the original work is properly cited, and is not altered, transformed, or built upon in any way. The terms on which this article has been published allow the posting of the Accepted Manuscript in a repository by the author(s) or with their consent.

between people, between things, and between people and things. According to Simondon, technology is a network of relationships. The neuroscientific study of relationships and social practices mediated by digital technology thus offers a new and indispensable perspective on intersubjectivity and the self.

That said, today we are witnessing something new: new technologies and digital media, together with the advent of the web 2.0, have radically changed the impact of images, sounds and words on our societies. This epochal change intercepts multiple aspects of individual existence, likely conditioning not only its development and adaptive capacities, but also acting on the sense of reality. If today we want to address the interpsychic dimension of human existence, we must carefully consider digital connectivity and its growing impact on our lives. It is important to underline, as noted by the anthropologist John Thompson, that '*The media should not be analyzed in isolation, in terms of intrinsic properties, […], but should be analyzed in relation to the forms of action and interaction that the use of the media brings into being*' (Thompson, 2020, 4). Embodied cognition offers an ideal vantage point to study how digital technologies impact on selves and their social relations for at least two reasons: first, because of the bodily performative character of the relations and interactions these new media evoke; second, because similar brain-body mechanisms ground our relations with the physical world and with its digital mediations (Gallese 2020a, 2020b; Gallese & Morelli, 2024).

In this article I first concisely address social cognition from an embodied perspective, introducing the notion of the bodily self and its social nature. Then I address how the impact of new digital technologies might affect the way we relate to the world and to others.

Embodied interaction: a foundation for social cognition

Neuroscientific research in the last thirty years has redefined the themes of the subject and intersubjectivity and, in more general terms, has given us a new conception of the so-called human mind, putting the body, relations and affectivity at the center (Gallese & Morelli, 2024). One of the most interesting contributions of neuroscience has been the discovery of the cognitive role of the cortical motor system. The teleological dimension of behavior, made up of actions aimed at achieving goals, is an integral part of the functional properties of the motor system, in close coupling with the brain centers that mediate appetitive and defensive behaviors, reward and emotions. Another aspect of the cognitive role of the motor system is represented by its involvement in perception: premotor and parietal areas contain motor neurons that also respond to visual, auditory and tactile inputs, in the absence of any movement. In this 'perceptual' mode the cortical motor system is partly active, but the movement is not produced: it is only simulated (for a review, see Gallese, 2009, 2014; Gallese & Guerra, 2019).

The bodily self, deeply ingrained in our sensorimotor experiences, constitutes a fundamental aspect of our understanding of selfhood (Blanke & Metzinger, 2009; Damasio, 2000; Gallese, 2001). This conceptualization emphasizes the intricate relationship between bodily awareness and environmental interaction, as highlighted by Merleau-Ponty (1945) and further elaborated by Gallese and Sinigaglia (2010, 2011). Central to this perspective is the role of the motor system in shaping our sense of agency and ownership over bodily actions (Gallese & Sinigaglia, 2010; Legrand, 2006). As such, the motor system serves as a foundational mechanism that integrates sensory information with our bodily experiences, defining the boundaries of our bodily self.

Expanding upon this notion, the connection between the motor system and the bodily self underscores the dual function of the latter: it not only constitutes essential aspects of selfhood but also informs our perception of others as embodied selves with similar motor capabilities (Gallese & Sinigaglia, 2010). This theoretical framework suggests that our understanding of selfhood is intricately linked to our embodied interactions with the world around us.

The discovery of mirror neurons, first in macaques and then in humans (Gallese, 2014; Gallese et al., 1996; Rizzolatti et al., 1996), has revealed the cognitive role of the motor system in social cognition, showing a neurophysiological mechanism that maps the agentive relationship between individuals (for a recent review, see Bonini et al., 2022). Mirror neurons spell out intersubjectivity first as intercorporeality. The ability to recognize others as intentional agents does not depend exclusively on linguistic competence but is primarily linked to the relational nature of action. Thus, intercorporeality becomes the primary source of knowledge of others. Empirical research has shown that the human brain is also equipped with mirroring mechanisms in the domain of emotions and sensations: the same brain regions involved in the subjective experience of emotions and sensations are also active when the same emotions and sensations are recognized in others (Gallese, 2007, 2014). Through embodied simulation, we engage with others

in social interactions while maintaining the distinction between self and other (Ferri et al., 2011).

My theory of embodied simulation (Gallese, 2003, 2005, 2007, 2014) proposes that there are mental states or processes that are corporeal because of their corporeal non-linguistic format of representation/modeling, at the sub-personal neurophysiological level. The corporeal format of a mental representation limits what it can represent and is in turn limited by the specific constraints of the human body that condition its behavior. Similar constraints apply to both one's own actions, emotions, or sensations, and to those of others. These constraints, common to all individuals, determine a common corporeal format of representation. Embodied simulation is the reuse of representations in corporeal format to map others. The nature and range of what can be recognized and understood with embodied simulation is limited by the corporeal format of the representations involved.

For us humans, however, the physical world is just one of the many parallel worlds we inhabit. In fact, the history of humanity shows how important human-made images and stories have been in the development of communities and social institutions, feeding collectively shared hierarchies, habits, social practices, rituals, rules, and beliefs. This contributed in an essential way to determine the sense we humans entertain of being part of social communities characterized by shared practices, values and symbols (Gallese, 2018a, 2018b).

In this context, the body serves as the paradigm for knowledge acquisition through interactions with the physical world, shaping our social bodily self (Gallese, 2014; Gallese & Cuccio, 2015). This perspective challenges the traditional dichotomy between individual and social cognition, suggesting that human cognition is inherently social and deeply embodied (Gallese, 2003). As we live in a new social landscape shaped by digital technologies, understanding the ways in which bodily selves interact in this digital realm becomes crucial.

Considering what we already know about the similarity between the brain-body mechanisms enabling our relationship with the world of objects and the world of others and their analogic and digital representational mediations, in the next sections I'll explore the current mediascape, generational effects in digital media use, and the social psychological, cultural, and public health correlates or sequels (or, perhaps, dangers) of patterns of digital media use, connecting neuroscientific empirical evidence to Benjamin's concept of 'innervation'.

The contemporary mediascape and its cognitive impact

The term 'mediascape' was introduced by the anthropologist Arjun Appadurai (1996) to describe both the distribution of the electronic devices producing and disseminating information, and the images of the world created by these media. Francesco Casetti more recently defined the mediascape as '*an environment that promotes or facilitates mediation between individuals and reality, thanks to technological artifacts placed in it*' (Casetti, 2018, p. 25). Drawing on Walter Benjamin's concept of 'innervation,' Casetti argued that media 'innervate' the territory by fostering new sensibilities that refine and update our perception of things and events.

The advent of smartphones and the parallel development of social media and its subsidiary applications have shifted the balance toward a mediated and represented reality, at the expense of factual presence and performative relationships with the physical bodies of others. The pervasiveness of the contemporary digital mediascape undeniably shapes our relationship with the world, reshaping the very concept of 'reality'. This influence can be analyzed from various perspectives, focusing on the anthropological, economic, or political aspects of social life. Here, the focus is primarily on the cognitive impact of our increasing immersion in digital screens.

As established earlier, human existence is characterized by the development of cognitive technologies that enhance our adaptation to the environment. However, contemporary digital media may go beyond this function, potentially becoming the primary mode of daily interaction with the world. This is facilitated by the ubiquitous use of mobile devices, granting access to a vast array of services, news feeds, social interactions, navigation tools, education, and entertainment.

Indeed, a noteworthy aspect of the contemporary mediascape is the progressive blurring of the line between 'physical reality' and its digital representation. This is particularly salient considering the shared neurophysiological and bodily mechanisms underlying the experience of the real world and its mediated fictional representations. Shortly, empirical evidence shows that some of the sensorimotor mechanisms enabling our daily relationships to physically present objects and other living beings also similarly operate when relating with both their analogic and digital representations. This applies to cultural artifacts, like paintings, sculptures and films but also to audio-visual content covering real events, like news feed or digitally-mediated social interactions, like video-chatting on social media

(see Gallese, 2018 a, 2018b, 2020a, 2020b; Gallese & Guerra, 2019).

Of course, the overlap is only partial, otherwise we would not be able to tell the difference between physical and digital presence. However, the point I want to make is that there are no radically distinct brain-body mechanisms to support our life in the physical world and that lived in the digital mediasphere. Given the high plasticity of brain mechanisms, it is an open question whether and to which extent the ever-growing time spent in the mediasphere will affect our 'analogic' life and interpersonal relationships.

The pervasiveness of digital media in social interactions raises questions about intersubjectivity and its dependence on embodied practices. Social media platforms multiply and potentially replicate these practices, creating a two-dimensional world reminiscent of fictional representations. This phenomenon can be viewed through a psychoanalytic lens. Paraphrasing Freud (1907), the repetitive nature of social media use might be seen as the transformation of personal anxieties into a 'religion' with high anxiety content, albeit a less private one due to its public nature. Social media use indeed displays some common traits with religion, like shared and repetitive practices and their ritualization. The COVID-19 pandemic has significantly amplified this trend. Public health measures limiting physical proximity have driven social life and relationships even more online.

The digital media environment and the sense of reality

The new digital technologies and media of the 21st century seem to challenge the basic definition of what is 'real'. Already about thirty years ago, scholars like Guy Debord and Jean Baudrillard pointed out that contemporary societies have progressively transformed themselves into their representations, losing authenticity. According to Debord, '*Everything that was once lived directly is now simply represented at a distance. The spectacle is not a collection of images; it is a social relationship between people mediated by images*' (2002, p. 7). And again, '*When the real world is transformed into simple images, simple images become real beings - dynamic inventions that provide the direct motivations for hypnotic behavior*' (2002, p. 9).

The sociologist and philosopher Jean Baudrillard has argued that contemporary society lives in a hyper-real world, where the model becomes reality without reality and where the *Simulacrum*—the copy of the real without the real—becomes itself reality (see Smith, 2010). As Baudrillard wrote: '*Let's be clear about this: if the Real is disappearing, it is not for lack of it - on the contrary, there is too much of it. It is the excess of reality that puts an end to reality, just as the excess of information puts an end to information, or the excess of communication puts an end to communication*' (Baudrillard, 2000, pp. 65–66).

These ideas resonate with Alan Badiou's (2018) exploration of the 'real' as both a constraint and a potential source of human emancipation. He ponders the question of whether human activity can transform the real, creating new possibilities for freedom.

The potentially dystopian impact of new media and new technologies, already predicted and theoretically discussed almost 30 years ago by postmodern thinkers, requires a new approach, since new media and their use have introduced radical changes in the way people perceive reality and live their lives, through the bodily changes introduced by new individual and social practices. It is necessary to focus on these changes and on how they in turn modify people's attitudes, feelings, opinions, and emotions. In short, how people's life experience changes. Let us see how and why this is important for the theme of the self and its interpersonal relationships.

The psychic implications of the transfer of psycho-emotional investment from the 3D physical world to the digital world are still poorly explored and studied today. Facial mimicry and emotional experience normally accompany real social interactions. A recent study (Hsu et al., 2020) measured subjective ratings of valence and arousal and facial electromyography (EMG) activity in the *zygomaticus major* and *corrugator supercilii* muscles of individuals, comparing live video interactions with pre-recorded videos, both showing positive (smiling) and negative (frowning) facial expressions. Subjective ratings showed that the live facial expressions elicited higher valence and more arousing than the corresponding pre-recorded videos for positive emotion conditions. Facial EMG data showed that compared with the pre-recorded video, live facial expressions more effectively elicited facial muscular activity congruent with the models' positive facial expressions. These findings indicate that emotional facial expressions in digitally mediated live social interactions are more evocative of emotional reactions and facial mimicry than earlier experimental data have suggested.

This suggests that perhaps we must revise and/or broaden both our notions of 'real' and 'presence', as it appears that from the time of the cave paintings of Chauvet onwards, humans constantly create and

inhabit a multiplicity of parallel worlds the experience of which is equally vivid, meaningful and important.

From selfies to deepfakes

It is fair to say that digital technologies, particularly portable devices, have determined a multifaceted impact on the construction of the self-image. The widespread adoption of smartphones has popularized self-portraits, or 'selfies'. As Giovanni Stanghellini recently wrote on the selfie, '…*one could speak of a pornographic view of the self, where feeling (and pleasure) comes through seeing one's own (and another's) body in action reflected as in a mirror, rather than immediately, i.e., coenesthetically, perceiving one's own body (and in one's flesh the flesh and pleasure of the other).* […] *The selfie is a symptom of the evanescence of the flesh, which it attempts to make up for through the visibility of the body*' (Stanghellini, 2020, pp. 62, 98).

The myth of Narcissus speaks of the deadly outcome of any gaze that fails to step outside the self, that fails to engage the other in an exchange of expressions, affects and meaning. Narcissism cancels the other, keeping the self within a bubble of self-referentiality. As Vittorio Lingiardi (2021, p. 178) recently wrote, '*an acrobat of self-esteem, Narcissus walks the tightrope between healthy self-love and his pathological self-aggrandizement. Between these extremes lies the narcissism of our daily lives, conditioned by the cultural context and definitely on the rise*.' Now, is this what characterizes the digital practice of the selfie? Empirical studies on this matter are inconclusive: while some authors link the practice of the selfie to narcissism (Lee & Sung, 2016; Nash et al., 2019), others highlight the limitations of establishing causality (Halpern et al., 2016). These latter authors in their longitudinal study suggest a reciprocal relationship, where narcissists take more selfies, and increased selfie-taking reinforces narcissism.

Undeniably, the selfie phenomenon has fueled the global rise of aesthetic surgery. Dissatisfied with the limitations of digital filters offered by apps like Face Tune, individuals increasingly turn to plastic surgeons to permanently alter their appearance. The goal is often to approximate the standardized, celebrity-driven beauty ideals promoted on platforms like Instagram and TikTok. As Tolentino (2019) aptly observed, these platforms prioritize 'good content'—content that is shared, replicated, and generates virality. Much of this content revolves around facial aesthetics, fostering the notion that beauty translates to social capital. Consequently, a growing trend sees young people modifying their bodies to mimic these idealized models, effectively transforming themselves into real-life embodiments of digitally constructed beauty. Perhaps it is not coincidental that narcissism is so highly present in the contemporary debate.

Further blurring the lines between reality and artifice are facial recognition technologies like Facebook's Deep Face. This deep learning system utilizes a 9-layer neural network to identify faces in digital images. Trained on millions of user-uploaded photos (often without explicit consent), Deep Face facilitates tagging in social media but also raises privacy concerns. When combined with artificial intelligence and Big Data, facial recognition tools can be used for biometric surveillance, potentially transforming social interaction into a highly monitored aesthetic regime. Ubiquitous CCTV cameras and webcams contribute to this ever-present gaze, blurring the line between science fiction and reality.

The future of advertising is already replacing human models with Deepfakes – synthetic media generated by deep learning algorithms like Generative Adversarial Networks (GANs). Deepfakes can produce realistic, fabricated images of people who don't exist. These algorithms are rapidly improving, potentially rendering the distinction between real and fake faces increasingly difficult. Applications include reviving deceased actors in films and enhancing de-aging techniques in movies. However, the most alarming application lies in creating fake video footage of public figures. This underscores the potential threats deepfakes pose to democratic institutions. Research is urgently needed to equip individuals with the skills to discern real from fake, exemplified by online games like 'Which Face is Real?' developed by West and Bergstrom (2019).

Exploring the aestheticization of social life and politics in the digital age

The ubiquity of smartphone cameras allows constant documentation of daily life—places visited, people encountered, meals consumed—and their subsequent sharing on social media. This incessant exchange of digital testimonies, driven by the desire for approval, has reshaped the fabric of social interaction and self-expression (Turkle, 2011, 2015). Life experiences, once lived and remembered, now seem validated only when recorded and shared, transforming life into a series of fleeting 'peak scenes' consumed and discarded at an accelerating pace.

Digital technologies facilitate the construction of a meticulously curated self-image, previously impossible.

The shared digital self-image composed of innumerous posts becomes the main referent when relating to the world of others, adding to the self a level of narcissistic self-reflection, even when not indulging in taking selfies. News, entertainment, and personal experiences converge on the same digital platforms, shaping users' attitudes towards communication styles.

While traditional media outlets have long influenced public discourse, the advent of digital media has democratized access to information and expanded the reach of individuals' voices (Castells, 2009). However, this democratization has also led to the proliferation of misinformation and the blurring of lines between fact and fiction (Wardle & Derakhshan, 2017). Furthermore, the phenomenon of the 'digital bubble' complicates the issue by contributing to the atomization of societies, where individuals rely more on mediated digital interactions than on face-to-face relationships (Sunstein, 2001). While digital bubbles may facilitate the incorporation of diverse perspectives through shared attention (Deroy, 2019), they also risk reinforcing echo chambers where individuals are exposed only to information that aligns with their pre-existing beliefs. The implications of these trends for our understanding of society and societal cohesion are profound.

Political discourse has undergone a significant transformation in the digital era. The rise of social media platforms has enabled politicians to directly engage with voters, bypassing traditional media gatekeepers and crafting carefully curated personas online (Bennett & Segerberg, 2013). Platforms such as Facebook, Instagram, and X have become instrumental in shaping political communication strategies, with politicians leveraging various forms of multimedia content to cultivate voter identification (Kreiss, 2016). Images of politicians in casual settings, strategic deployment of emotionally charged language, and use of simplified slogans all contribute to the 'aestheticization' (see Lipovetsky & Serroy, 2013) of political discourse. While the study of these techniques in political discourse has a long history (Lakoff, 1996, 2004, 2008; Westen, 2007), the amplification power of social media is unprecedented. Simplified slogans and outright falsehoods can now spread virally, reaching millions instantaneously. The sharing of memes likely homogenizes reactions and preferences, attuning millions to the same topics and opinions. Kramer et al. (2014) further confirm this through experiments showing emotional contagion on social media platforms. These findings suggest that the digital sphere, by maximizing content diffusion, may synchronize users' brains, particularly for highly arousing content. This has significant implications for political consensus building, warranting further empirical investigation.

The advent of modern media has introduced a concerning capability to distort perceptions of reality, potentially leading individuals to accept falsehoods as truth. Traditional news outlets, including radio and television, have long been regarded as reliable sources of information. However, the landscape of media consumption has evolved significantly. Television, once the predominant medium, has been supplanted by multiple digital devices such as computers, tablets, and mobile phones, hence being 'remediated'. Bolter and Grusin (2000) argued that media no longer function in isolated spheres but are interconnected and remediated across various platforms, so that '*no medium … can now function independently and establish its own separate and purified space*' (2000, p. 55).

The dissemination of fake news and the manipulation of emotional responses on social media platforms have raised concerns about the erosion of trust in institutions and the polarization of public discourse (Allcott & Gentzkow, 2017). Already 6 years ago, the journal Science dedicated its cover to fake news, inspired by the research article by Vosoughi et al. (2018), showing that fake news diffuses significantly farther, faster, deeper, and more broadly than the truth in all categories of information, the more pronounced effects being for false political news. The same study revealed that robots accelerated the spread of true and false news at the same rate, implying that false news spreads more than the truth because humans, not robots, are more likely to spread it. The same authors argued that people prefer fake news because they consider them more novel and unexpected than true ones, and while the latter inspire anticipation, sadness, joy and trust, the former evoke surprise, fear and disgust, likely boosting people's arousal, anxiety and anger. Another study showed that young individuals find difficult to distinguish neutral sources from biased ones, and fake accounts from real ones (McGrew et al., 2018). As the authors write, '*Although the Internet has the potential to democratize access to information, it puts enormous responsibility on individuals to evaluate the reliability of information. If citizens are not prepared to critically evaluate the information that bombards them online, they are apt to be duped by false claims and misleading arguments*' (2018, p. 155).

Pennycook et al. (2018) demonstrated that exposure to fake news leads individuals to inaccurately perceive it as more credible—a phenomenon termed the 'illusory truth effect.' This effect persists even when the fake news is flagged as contested by fact-checkers or

contradicts the reader's political beliefs. The authors argue that social media platforms contribute to the dissemination of false narratives, and simply labeling stories as disputed is insufficient to counteract this trend. Additionally, research by Zollo et al. (2015) indicates that prolonged discussions on social media about news topics tend to foster increasingly negative sentiments among commentators: the longer the discussion, the greater the negativity of the sentiment. As the number of comments increases -that is, as the discussion becomes longer- the emotional color of the post becomes increasingly negative.

In response to these challenges, innovative approaches are needed to promote media literacy and critical thinking skills among the public (Wardle & Derakhshan, 2017). Additionally, interdisciplinary research that integrates insights from neuroscience, psychology, and communication studies can provide valuable perspectives on the cognitive and emotional mechanisms underlying individuals' responses to digital media (Roozenbeek & van der Linden, 2019).

'Digital natives' and mental well-being

Throughout history, the emergence of novel technologies altering the representation and reproduction of the world has consistently sparked harsh criticisms and predictions of impending doom. From Plato onwards, various cognitive technologies such as literacy, poetry, photography, cinema, television, the internet, and smartphones have been vilified for their perceived negative impacts on individuals and society. However, these new cognitive technologies have progressively broadened human capacities for creative expression. While some may be inclined to dismiss concerns surrounding the adoption of digital technologies as conventional anxieties stemming from conservative attitudes towards progress and innovation, it is nonetheless difficult to ignore the potential for unprecedented changes in our lives and societies brought about by the advent of these new digital technologies.

Indeed, the present public debate on the digital mediasphere and its impact on human life and human societies, particularly after the recent booming of AI, is somehow polarized between acritical techno-enthusiasm on the one hand, and the praising and longing for the good old analogic times, on the other. This polarization becomes even more evident when discussing the relationship between the pervasive use of digital media and mental well-being, particularly in the case of the youngest generations, the so-called 'digital natives'.

On average, it is fair to say that in the current public debate the prevalent attitude towards digital technologies in relation to mental health is negative, particularly for the younger generations. A recent book by the American psychologist Jonathan Haidt, *The Anxious Generation* (2024), devoted to the analysis of the effect of widespread early use of digital technologies among generation Z individuals (people born after 1995), epitomizes this negative attitude with sentences like the following: '*Gen Z became the first generation in history to go through puberty with a portal in their pockets that called them away from the people nearby and into an alternative universe that was exciting, addictive, unstable, and—as I will show—unsuitable for children and adolescents*' (p. 19); '*The first generation of Americans who went through puberty with smartphones (and the entire internet) in their hands became more anxious, depressed, self-harming, and suicidal*' (p. 74).

This almost apocalyptic view is apparently based on empirical research showing an abrupt and steady increase of self-harming and suicidal episodes among adolescents, starting from year 2010, when Apple introduced the frontal camera in the iPhone and when social media reportedly took over. According to Haidt, by displacing physical play and in-person socializing, smartphones and social media have rewired childhood and changed human development on an almost unimaginable scale.

Since big claims require big evidence, where are we? Arguing that digital-technology use and adolescent mental-health problems are rising together is merely correlative, at best, with no hints for a real causal relationship between the two phenomena. In a recent review of this book that appeared in Nature last March, Candice Odgers writes: '*Hundreds of researchers, myself included, have searched for the kind of large effects suggested by Haidt. Our efforts have produced a mix of no, small and mixed associations. Most data are correlative. When associations over time are found, they suggest not that social media use predicts or causes depression, but that young people who already have mental-health problems use such platforms more often or in different ways from their healthy peers*' (2004, p. 29). Indeed, several meta-analyses and systematic reviews converge on the same message. Odgers reports that the Adolescent Brain Cognitive Development study, the largest long-term study of adolescent brain development in the United States, has found no evidence of drastic changes associated with digital technology use.

Orben and colleagues (2019) at Oxford University published a study concerning a large cohort of 12,672

young individuals 10–15 years-old, showing that social media use is not, in and of itself, a strong predictor of life satisfaction across the adolescent population. One of the biggest meta-analytic studies on this topic was published by Jeff Hancock et al. (2022) at the Media Lab of Stanford University. It concerned all the available scientific studies ($n=226$) published since 2006 on the relationship between the amount of time spent on social media and psychological well-being in young individuals ($n=275{,}728$). The results showed very small significant effects, with a trade-off between increased depression and anxiety, along with improved social well-being.

Another large study carried out in Norway recently investigated the relationship among social media use, social skills and off-line friendships in a cohort of 1007 individuals of 10–18 years-old (Steinsbekk et al., 2024). The results showed that increased social media use predicted more time with friends offline but was unrelated to future changes in social skills. Age and sex did not moderate these associations; increased social media use predicted declined social skills only among adolescents high in social anxiety symptoms. The authors of this study conclude that social media use may neither harm nor benefit the development of social skills and may promote, rather than displace, offline interaction with friends during adolescence. However, increased social media use may pose a risk for reduced social skills in socially anxious individuals.

It is true that suicide rates have been increasing steadily for the past 20 years in most western countries and that suicide, self-harm episodes, depression and anxiety have been rising particularly among adolescents. Indeed, Twenge et al. (2018) proposed a causal relationship with screen-time. Other researchers, however, cite access to guns, exposure to violence, structural discrimination and racism, sexism and sexual abuse, the opioid epidemic, economic hardship and social isolation as leading contributing factors (see Martínez-Alés et al., 2022). At this stage the better strategy would be to have more large-scale longitudinal well-controlled studies.

Conclusions

Science should not be prescriptive, as its main role is to shed light on phenomena rather than judging them. The contemporary debate on digital technologies too often opts for prejudices and stereotyped judgments, be they acritically negative or positive, instead of relying on what can be empirically ascertained by research. As we have seen throughout the article, we still know very little on how digital media affect selves and their social relationships: we just need more research. Furthermore, digital technologies are only one aspect characterizing contemporary western societies. Before blaming digital technology, we should perhaps consider the fact that its specific use is dictated and conditioned by the contemporary model of society in which the same technology is produced and used. Our western capitalist model speaks more of individuals than of groups, praises self-empowerment and competition at the expense of solidarity, leading people to an ever-growing anxious fight for self-promotion. The declining birth rate increasingly produces families with only children, hence developing fearful and overprotective attitudes at the expense of developing their children's sense of autonomy and responsibility. To this must be added a political discourse that for mere electoral purposes often promotes a frustrating sense of fear and anxiety among citizens, cynically exploited to gain wider consensus and consolidate power. All these factors add up and most likely affect people's mental health.

We should also bear in mind that unsettling views on digital media could be complemented by more hopeful scenarios that leave ample room to the creative development of new digitally mediated ways of living. Carbone and Lingua recently wrote, '*the very real forms of confessionality adopted on screens can also promote practices of subjectivation that allow us to discursively appropriate our identities*. […] *In the storytelling -verbal and iconic at once- practiced on social media, users thus highlight a real need to establish social ties, and their experience of fragmented individuality proves functional in constructing new identities that compose and recompose the relationships made possible by digital environment*' (2023, pp. 168, 173).

As these words suggest, we have entered a new dimension of sociality that brings along new forms of subjectivation and sociality, hence we should expect, particularly for younger generations, a radical change for good and for bad in the quality and quantity of interpersonal relations entertained in daily lives. It is an easy bet to envision that this will have -according to many already has- strong implications for psychiatry too. If we allow a view of psychopathology in terms of 'interpersonal misattunement' (see Bolis et al., 2023), then the new forms of social interactions fostered by the digital mediascape should be more and more in the focus of empirical research, where social neuroscience can provide an important contribution.

As we explore the complexities of the digital age, it is imperative to critically examine the role of digital technologies in shaping social life and political discourse. By understanding the interplay between

content, emotional context, delivery methods, and shareability within digital media landscapes, we can develop strategies to mitigate the negative effects of misinformation and promote informed decision-making in our increasingly digital world.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

This work was funded by the Ministry of University and Research (MUR), National Recovery and Resilience Plan (NRRP) Project MNESYS [PE00000006] – A Multiscale Integrated Approach to the Study of the Nervous System in Health and Disease [DN 1533 11.10. 2022] to V.G. and by the PRIN Grant 2020YB7J25 to V.G.

References

Allcott, H., Gentzkow, M. (2017). Social Media and Fake News in the 2016 Election. *Journal of Economic Perspectives*, *31*(2), 211–236.

Appadurai, A. (1996). *Modernity at large: Cultural dimensions of globalization*. University of Minnesota Press.

Badiou, A. (2018). Badiou and *his interlocutors. Lectures, interviews and responses* (A. J. Bartlett & J. Clemens, Eds.). Bloomsbury.

Baudrillard, J. (2000). *The vital illusion* (pp. 65–66). Columbia University Press.

Benjamin, W. (2002). *Selected writings. Volume 3 (1935–1938)* (H. Eilan & M. W. Jennings, Eds.). The Belknap Press of Harvard University Press.

Benjamin, W. (2008). *The work of art in the age of mechanical reproduction* (J. A. Underwood, Trans.). Penguin Books.

Bennett, W. L., & Segerberg, A. (2013). The logic of connective action: Digital media and the personalization of contentious politics. *Information, Communication & Society*, *16*(5), 1–30.

Blanke, O., & Metzinger, T. (2009). Full- body illusions and minimal phenomenal selfhood. *Trends in Cognitive Sciences*, *13*(1), 7–13. https://doi.org/10.1016/j.tics.2008.10.003

Bolis, D., Dumas, G., & Schilbach, L. (2023). Interpersonal attunement in social interactions: From collective psychophysiology to inter-personalized psychiatry and beyond. *Philosophical Transactions of the Royal Society of London. Series B, Biological Sciences*, *378*(1870), 20210365. https://doi.org/10.1098/rstb.2021.0365

Bonini, L., Rotunno, C., Arcuri, E., & Gallese, V. (2022). Mirror neurons 30 years later: Implications and applications. *Trends in Cognitive Sciences*, *26*(9), 767–781. https://doi.org/10.1016/j.tics.2022.06.003

Bolter, J., & Grusin, R. (2000). *Remediation: Understanding new media*. MIT Press.

Carbone, M., & Lingua, G. (2023). *Towards an anthropology of screens. Showing and hiding, exposing and protecting*. Pallgrave Macmillan.

Casetti, F. (2018). Mediascape. A decalogue. In S. de Silva, D. Furioso, & S. Jaff (Eds.), *Perspecta 51, medium* (pp. 21–43). MIT Press.

Castells, M. (2009). *Communication power*. Oxford University Press.

Damasio, A. (2000). *The Feeling of What Happens: Body and emotion in the making of consciousness*. Harvest Books.

Debord, G. (2002). *The society of the spectacle*. Hobgoblin Press.

Deroy, O. (2019). *The Danger of Bursting Bubbles. Why informational bubbles are a necessary part of our world. Changing How the World Thinks. An online magazine of big ideas.* Issue 83. https://iai.tv/articles/the-danger-of-bursting-bubbles-auid-1281#

Ferri, F., Frassinetti, F., Costantini, M., & Gallese, V. (2011). Motor simulation and the bodily self. *PloS One*, *6*(3), e17927. https://doi.org/10.1371/journal.pone.0017927

Freud, S. (1907). Obsessive acts and religious practices. In J. Strachey (Ed.), *The standard edition of the complete works of Sigmund Freud*. Hogarth Press.

Gallese, V. (2001). The "Shared Manifold" Hypothesis: From mirror neurons to empathy. *Journal of Consciousness Studies*, *8*(5–7), 33–50.

Gallese, V. (2003). The manifold nature of interpersonal relations: The quest for a common mechanism. *Philosophical Transactions of the Royal Society of London. Series B, Biological Sciences*, *358*(1431), 517–528. https://doi.org/10.1098/rstb.2002.1234

Gallese, V. (2005). Embodied simulation: From neurons to phenomenal experience. *Phenomenology and the Cognitive Sciences*, *4*(1), 23–48. https://doi.org/10.1007/s11097-005-4737-z

Gallese, V. (2007). Before and below Theory of Mind: Embodied simulation and the neural correlates of social cognition. *Philosophical Transactions of the Royal Society of London. Series B, Biological Sciences*, *362*(1480), 659–669. https://doi.org/10.1098/rstb.2006.2002

Gallese, V. (2009). Motor abstraction: A neuroscientific account of how action goals and intentions are mapped and understood. *Psychological Research*, *73*(4), 486–498. https://doi.org/10.1007/s00426-009-0232-4

Gallese, V. (2014). Bodily Selves in Relation: Embodied simulation as second-person perspective on intersubjectivity. *Philosophical Transactions of the Royal Society of London. Series B, Biological Sciences*, *369*(1644), 20130177. https://doi.org/10.1098/rstb.2013.0177

Gallese, V. (2018a). The power of images: A view from the brain-body. *Phenomenology and Mind*, *14*, 70–79.

Gallese, V. (2018b). Naturalizing aesthetic experience. The role of (liberated) embodied simulation. *Projections*, *12*(2), 50–59. [Mismatch https://doi.org/10.3167/proj.2018.120207

Gallese, V. (2020a). Brain, body, habit, and the performative quality of aesthetic experience. In I. Testa & F. Caurana

(Eds.), *Habits: Pragmatist approaches from cognitive neuroscience to social science* (pp. 376–394). Cambridge University Press.

Gallese, V. (2020b). Aesthetic World. Experience and performativity in the digital era. *Reti, Saperi, Linguaggi 1/2020 a*, *9*(17), 55–84.

Gallese, V., Fadiga, L., Fogassi, L., & Rizzolatti, G. (1996). Action recognition in the premotor cortex. *Brain: A Journal of Neurology*, *119 (Pt 2)*(2), 593–609. https://doi.org/10.1093/brain/119.2.593

Gallese, V., & Sinigaglia, C. (2010). The bodily self as power for action. *Neuropsychologia*, *48*(3), 746–755. https://doi.org/10.1016/j.neuropsychologia.2009.09.038

Gallese, V., & Sinigaglia, C. (2011). How the body in action shapes the self. *Journal of Consciousness Studies*, *18*(7–8), 117–143.

Gallese, V., & Cuccio, V. (2015). The paradigmatic body. Embodied simulation, intersubjectivity and the bodily self. In T. Metzinger & J. M. Windt (Eds.), *Open MIND* (pp. 1–23). MIND Group.

Gallese, V., & Guerra, M. (2019). *The emphatic screen. Cinema and neuroscience*. Oxford University Press.

Gallese, V., & Morelli, U. (2024). *Cosa Significa Essere Umani? Corpo, Cervello e Relazione per Vivere nel Presente*. Raffaello Cortina Editore.

Haidt, J. (2024). *Anxious Generation How the Great Rewiring of Childhood Is Causing an Epidemic of Mental Illnes*. London: Penguin Publising House.

Hancock, J., Liu, S. X., Luo, M., & Mieczkowski, H. (2022). Psychological well-being and social media use: A meta-analysis of associations between social media use and depression, anxiety, loneliness, eudaimonic, hedonic and social well-being. *SSRN Electronic Journal*. https://ssrn.com/abstract=4053961; https://doi.org/10.2139/ssrn.4053961

Halpern, D., Valenzuela, S., & Katz, J. E. (2016). "Selfie-ists" or "Narci-selfiers"?: A cross-lagged panel analysis of selfie taking and narcissism. *Personality and Individual Differences*, *97*, 98–101. https://doi.org/10.1016/j.paid.2016.03.019

Hsu, C. T., Sato, W., & Yoshikawa, S. (2020). Enhanced emotional and motor responses to live versus videotaped dynamic facial expressions. *Scientific Reports*, *10*(1), 16825. https://doi.org/10.1038/s41598-020-73826-2

Kramer, A. D., Guillory, J. E., & Hancock, J. T. (2014). Experimental evidence of massive-scale emotional contagion through social networks. *Proceedings of the National Academy of Sciences of the United States of America*. *11*(24), 8788–8790.

Kreiss, D. (2016). Seizing the moment: The presidential campaigns' use of Twitter during the 2012 electoral cycle. *New Media & Society*, *18*(8), 1473–1490. https://doi.org/10.1177/1461444814562445

Lakoff, G. (1996). *Moral politics: What conservatives know that liberals don't*. University of Chicago Press.

Lakoff, G. (2004). *Don't think of an elephant: Know your values and frame the debate*. Green Publishing.

Lakoff, G. (2008). *The political mind: Why you can't understand 21st-century American politics with an 18th-century brain*. Viking.

Lee, J.-A., & Sung, Y. (2016). Hide-and-seek: Narcissism and "selfie" related behavior. *Cyberpsychology, Behavior and Social Networking*, *19*(5), 347–351. https://doi.org/10.1089/cyber.2015.0486

Legrand, D. (2006). The bodily self: The sensori- motor roots of pre- reflective self- consciousness. *Phenomenology and the Cognitive Sciences*, *5*(1), 89–118. https://doi.org/10.1007/s11097-005-9015-6

Lingiardi, V. (2021). *Arcipelago N. Variazioni sul Narcisismo*. Einaudi.

Lipovetsky, G., & Serroy, J. (2013). *L'Esthétisation du monde. Vivre à l'âge du Capitalisme Artiste*. Éditions Gallimard.

Martínez-Alés, G., Jiang, T., Keyes, K. M., & Gradus, J. L. (2022). The recent rise of suicide mortality in the United States. *Annual Review of Public Health*, *43*(1), 99–116. https://doi.org/10.1146/annurev-publhealth-051920-123206

McGrew, S., Breakstone, J., Ortega, T., Smith, M., & Wineburg, S. (2018). can students evaluate online sources? Learning from assessments of civic online reasoning. *Theory & Research in Social Education*, *46*(2), 165–193. https://doi.org/10.1080/00933104.2017.1416320

Merleau-Ponty, M. (1945). *Phénomenologie de la Perception*. Gallinard.

Nash, K., Johansson, A., & Yogeeswaran, K. (2019). Social media approval reduces emotional arousal for people high in narcissism: Electrophysiological evidence. *Frontiers in Human Neuroscience*, *13*, 292. https://doi.org/10.3389/fnhum.2019.00292

Odgers, C. S. (2024). The great rewiring unplugged. Is social media really behind an epidemic of teenage mental illness? *Nature*, *628*(8006), 29–30. https://doi.org/10.1038/d41586-024-00902-2

Orben, A., Dienlin, T., & Przybylski, A. K. (2019). Social media's enduring effect on adolescent life satisfaction. *Proceedings of the National Academy of Sciences of the United States of America*, *116*(21), 10226–10228. https://doi.org/10.1073/pnas.1902058116

Pennycook, G., Cannon, T. D., & Rand, D. G. (2018). Prior exposure increases perceived accuracy of fake news. *Journal of Experimental Psychology. General*, *147*(12), 1865–1880. https://doi.org/10.1037/xge0000465

Pinotti, A. (2018). *Costellazioni. Le parole di Walter Benjamin*. Einaudi.

Rizzolatti, G., Fadiga, L., Gallese, V., & Fogassi, L. (1996). Premotor cortex and the recognition of motor actions. *Brain Research. Cognitive Brain Research*, *3*, 131–141. https://doi.org/10.1016/0926-6410(95)00038-0

Roozenbeek, J., & van der Linden, S. (2019). Fake news game confers psychological resistance against online misinformation. *Pallgrave Communication*, *5*, 65. https://doi.org/10.1057/s41599-019-0279-9

Simondon, G. (2001). *Du Mode d'Existence des Objets Techniques*. Paris: Aubier.

Smith, R. G. (2010). *The Baudrillard dictionary*. Edinburgh University Press Ltd.

Stanghellini, G. (2020). *Selfie. Sentirsi nello Suardo dell'Altro*. Feltrinelli.

Steinsbekk, S., Bjørklund, O., Valkenburg, P., Nesi, J., & Wichstrøm, L. (2024). The new social landscape: Relationships among social media use, social skills, and offline friendships from age 10–18 years. *Computers in Human Behavior*, *156*, 108235.

Sunstein, C. R. (2001). *Republic.com*. Princeton University Press.

Thompson, J. B. (2020). Mediated interaction in the digital age. *Theory, Culture and Society*, *37*(1), 3–28. https://doi.org/10.1177/0263276418808592

Tolentino, J. (2019, September 29). How TikTok holds our attention. *The New Yorker*.

Turkle, S. (2011). *Alone together: Why we expect more from technology and less from each other*. Basic Books.

Turkle, S. (2015). *Reclaiming conversation: The power of talk in a digital age*. Penguin.

Twenge, J. M., Joiner, T. E., Rogers, M. L., & Martin, G. N. (2018). Increases in depressive symptoms, suicide-related outcomes, and suicide rates among U.S. adolescents after 2010 and links to increased new media screen time. *Clinical Psychological Science*, *6*(1), 3–17. https://doi.org/10.1177/2167702617723376

Vosoughi, S., Roy, D., & Aral, S. (2018). The spread of true and false news online. *Science (New York, N.Y.)*, *359*(6380), 1146–1151. https://doi.org/10.1126/science.aap9559

Wardle, C., & Derakhshan, H. (2017). *Information disorder: Toward an interdisciplinary framework for research and policy making*. Council of Europe report.

West, J., & Bergstrom, C. (2019). *Which face is real*? Online game. https://www.whichfaceisreal.com/index.php

Westen, D. (2007). *The political brain. The role of emotion in deciding the fate of the nation*. Publicaffairs.

Zollo, F., Novak, P. K., Del Vicario, M., Bessi, A., Mozetič, I., Scala, A., Caldarelli, G., & Quattrociocchi, W. (2015). Emotional dynamics in the age of misinformation. *PloS One*, *10*(9), e0138740. https://doi.org/10.1371/journal.pone.0138740